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SC4052/CE4052/CZ4052 Cloud Computing – Assignment 2

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Topic: Tutorial 3 Question 6 (i)

PageRank-Based Crawl Prioritization for AI Web Crawling

1. Introduction

This report implements a PageRank-based crawl prioritization demo for the “AI web crawling for training” option in Question 6. A small directed URL graph is constructed manually, PageRank scores are computed using NetworkX, and the top (k) URLs are selected as a baseline crawl strategy based on authority. This directly satisfies the requirement to rank URLs using precomputed PageRank scores.

This report further explains why high-PageRank pages may provide better training data for generative AI models: such pages are often more central, more widely referenced, and more likely to contain substantial informational content. However, this report also shows that PageRank alone is not enough for real crawling decisions.

To address that limitation, this report introduces a heuristic-enhanced strategy that combines PageRank with lightweight metadata related to crawl permission, domain trust, content richness, freshness, and low-value page patterns. The resulting policy better reflects the practical goals of an AI crawler by preferring pages that are both authoritative and suitable for training-data collection.

The code used in this report can be obtained at: https://github.com/ParrotG/SC4052_Assignment2

2. Construction of a Small Directed URL Graph and PageRank-Based Top-(k) Ranking

2.1 Graph construction

This report constructs a small directed web graph manually. Each node is a readable URL, and each directed edge represents a hyperlink from one page to another. The graph includes several different categories of pages in order to simulate realistic crawl prioritization decisions:

- research and academic pages
- open-data pages
- journal article pages
- news pages
- spam-like pages
- login pages
- search-result-style pages

The tutorial question asks for a small graph represented as URLs and their outlinks, and it also asks for a heuristic that prefers high-quality pages that permit crawling. A manually constructed graph provides enough control to

demonstrate both goals clearly.

The current implementation contains 17 URL nodes and directed hyperlink relations among them. In particular, several high-value pages such as research articles, lab pages, open-data resources, and journal articles are densely connected, while lower-value pages such as spam, login, and query-heavy search pages are either weakly connected or explicitly marked as unsuitable for crawling.

2.2 PageRank computation

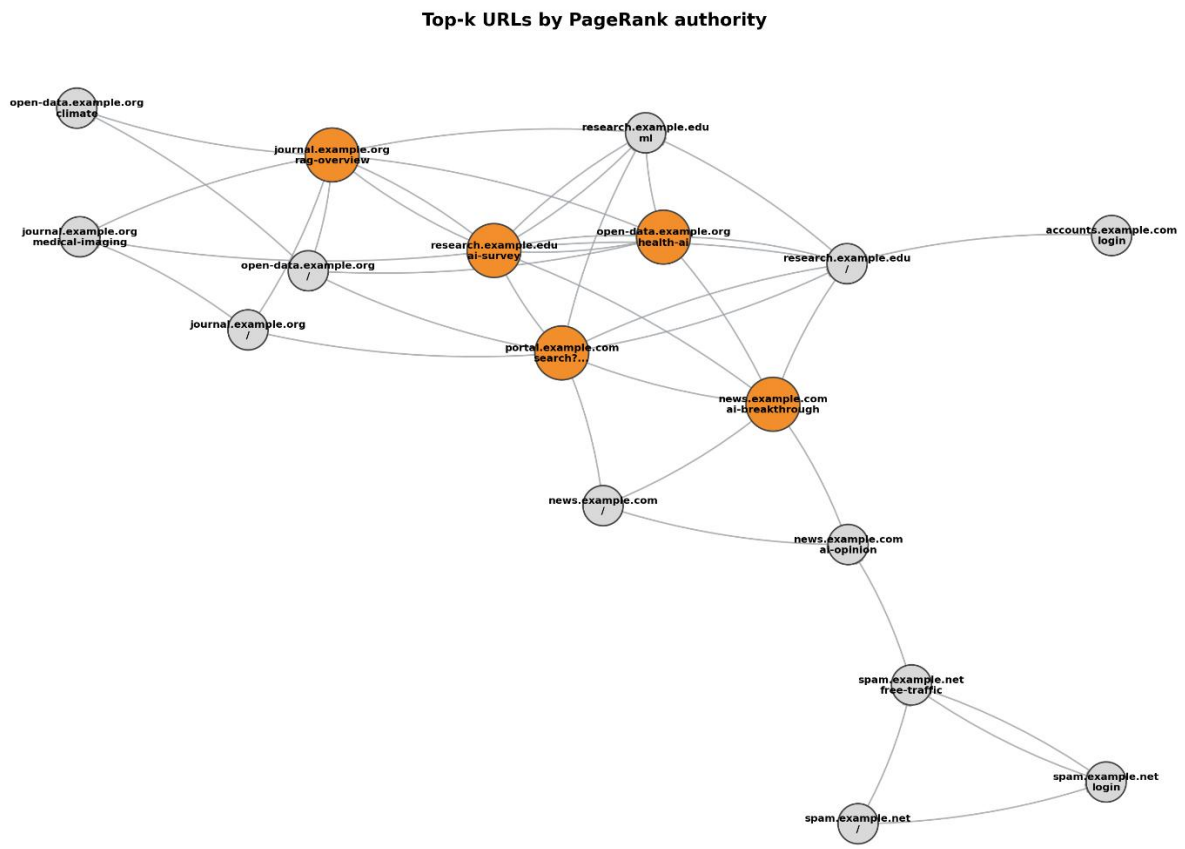
Since subtask assume work start with precomputed PageRank scores, this report uses the existing networkx.pagerank implementation with damping factor ($\alpha = 0.85$). The function compute_pagerank_scores takes the directed graph as input and returns a dictionary that maps each URL to its PageRank score.

2.3 Top-k ranking by authority

After obtaining PageRank scores, this report ranks all URLs in descending order of PageRank and returns the top-k results. In the implementation, this behavior is provided by get_top_k_by_authority, which sorts the (URL, score) pairs by PageRank and returns the highest-ranked entries. Optional command-line arguments allow the user to specify the desired value of k (by default, k=5).

This baseline strategy directly answers the first requirement of the assignment: given a directed web graph and precomputed PageRank scores, return the top-k URLs to crawl based on authority. It captures structural importance in the link graph without yet considering content quality or crawl permission constraints.

Figure 1 Top-k URLs highlighted under the pure PageRank authority strategy



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3. Why High-PageRank Pages May Provide Higher-Quality Training Data for Generative AI

High-PageRank webpages may provide higher-quality training data for generative AI models because PageRank reflects the authority and importance of a page in the web graph. Pages that are linked by many other important pages are often more trustworthy, better structured, and more information-dense. Such pages are more likely to contain accurate knowledge, clearer language, and representative content, which are valuable for training large language models. In addition, prioritizing high-PageRank pages allows an AI crawler to collect useful data more efficiently. However, PageRank alone is not sufficient, so it should be combined with other signals such as crawl permission, domain trustworthiness, and content quality.

4. Heuristic-Enhanced Crawl Prioritization

4.1 Motivation

The second part of the task asks for one heuristic to find high-quality pages that permit crawling. This report interprets that requirement as an enhancement to the baseline PageRank ranking. The baseline ranks pages only by authority; the enhanced policy keeps PageRank as the core signal but adjusts the crawl priority using lightweight metadata that approximates page quality and crawl suitability.

4.2 Metadata design

To support heuristic ranking, this report attaches a metadata record to each page. The implementation defines the following fields:

- allows_crawling: whether the page should be considered crawlable;
- content_length: whether the page is likely to contain substantial textual information;
- is_login_page and has_many_query_params: identify low-value utility pages;
- is_trusted_domain: rewards pages that resemble academic or non-commercial knowledge sources;
- freshness_days: provides preference for newer pages.

4.3 Heuristic scoring method

The implementation computes a heuristic-enhanced score for each page by starting from its PageRank score and then adding bonuses or penalties based on the metadata. If a page does not permit crawling, it is assigned a score of (-1.0), effectively removing it from consideration. The scoring logic then:

- adds a bonus for trusted domains,
- adds a bonus for larger content length,
- adds a bonus for freshness,
- subtracts a penalty for login-like pages,
- subtracts a penalty for pages with many query parameters, and
- subtracts penalty if the URL path suggests low-value page types such as login, signin, signup, account, search,

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tag, or filter.

This strategy keeps PageRank as the central authority signal while injecting a small amount of domain knowledge about which pages are more likely to produce useful training data.

4.4 Heuristic ranking results

The implementation produces a second top-(k) ranking using the heuristic-enhanced policy through the function get_top_k_with_heuristics. The script also generates a visualization that highlights the URLs selected by the enhanced strategy, allowing direct comparison with the pure PageRank baseline.

In general, this report expects the heuristic-enhanced strategy to preserve many of the highly authoritative academic and dataset pages while demoting or excluding pages that are unsuitable for training-data collection. For example, login pages and query-heavy search pages may still appear in the hyperlink graph and may even receive some structural importance, but they should not be prioritized for AI crawling because they either do not permit crawling or are unlikely to contribute useful language data. Similarly, spam-like pages may remain reachable in the graph but should be outranked by trusted, content-rich resources.
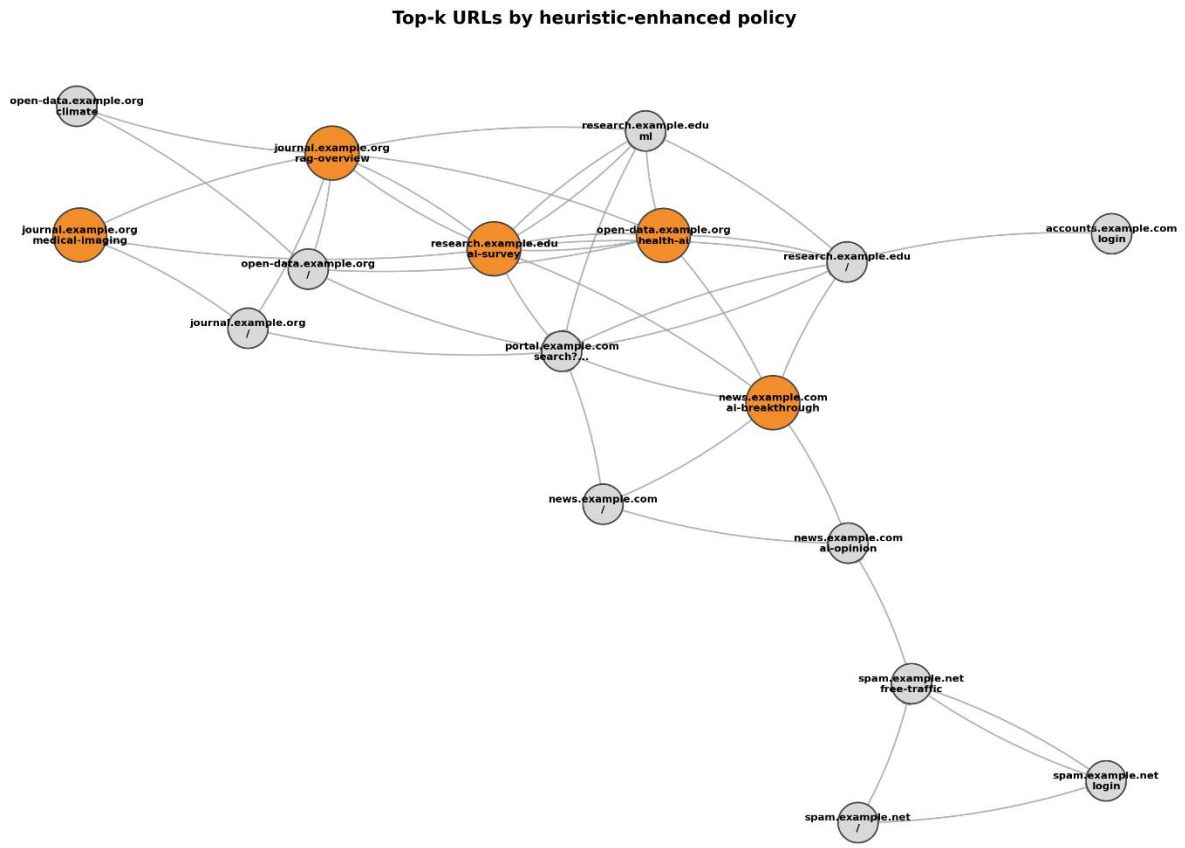
Figure 2 Top-k URLs highlighted under the heuristic-enhanced policy